DARSHAN MATHAPATI

Embedded Engineer

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LinkedIn GitHub IEEE Student Member

Career Objective

Electronics and Communication Engineering student with hands-on experience in embedded systems and IoT using ESP32, Raspberry Pi, and Arduino UNO. Skilled in sensor interfacing, cloud-connected monitoring, and hardware prototyping. Seeking an embedded engineering opportunity to build practical, real-world solutions while continuing to learn.

Education

East Point College of Engineering and Technology 2023 – 2027
B.E. in Electronics and Communication Engineering – 7.8 CGPA Bangalore, Karnataka

ICS Mahesh PU Science College 2021 – 2023
Karnataka State Pre-University Board – 87.66% Dharwad, Karnataka

Shri Sharada Vidyaniketan School 2020 – 2021
Karnataka Secondary Education Examination Board – 87.36% Vijayapura, Karnataka

Technical Skills

Programming: C/C++ (Arduino), Python (Raspberry Pi) – basics
Microcontrollers and Boards: ESP32, Arduino UNO, Raspberry Pi 4B
Communication Protocols: I2C, SPI, UART, Bluetooth (HC-05), Wi-Fi
Sensors and Modules: IR, Ultrasonic (HC-SR04), Gas, Fire, TDS, pH, Turbidity, Temperature, SIM800L (GSM), L298N Motor Driver, Servo Motors
Software and Tools: Arduino IDE, Thonny, Blynk IoT, Cadence, Overleaf (LaTeX)
ECE Tools: Proteus, Tinkercad, Wokwi, MATLAB, Raspberry Pi OS (Linux), ChatGPT, Claude AI

Hands-on Experience

- **Raspberry Pi 4B:** Camera interfacing, OCR-based text recognition, Excel data storage, voice alerts, and communication with ESP32.
- **ESP32:** IoT monitoring with Blynk Cloud, Wi-Fi/Bluetooth communication, mobile and email notifications, and servo control.
- **Arduino UNO:** Sensor interfacing, motor control using L298N, and Bluetooth-controlled robotics.
- **Hardware:** Breadboard prototyping, soldering, multimeter-based testing and debugging, and power regulation using buck converters.

Projects

Sparsh-Vaani: Medicine Label Reader for Differently-Abled Patients *Raspberry Pi, ESP32, OCR*
Built an assistive system that reads medicine labels using a camera and OCR, stores details in Excel, and controls a six-compartment pill dispenser through ESP32, with voice alerts, reminders, and IoT notifications.

Smart PillCare: IoT Medicine Dispenser for Elderly *ESP32, Blynk IoT*
Developed an automatic dispenser with pre-alert reminders, pill dispensing, remaining-tablet count, Blynk app notifications, and email alerts to caregivers for missed doses; documented in an IEEE-format paper.

AquaVision: Smart Water Quality Monitoring *ESP32, Blynk Cloud*
Built an IoT system to monitor TDS, pH, turbidity, and temperature, measure water level using HC-SR04, and send mobile notifications and buzzer alerts.

Disaster Management Robot *Arduino UNO, ESP32*
Designed a rescue robot with fire and gas detection, obstacle avoidance, Bluetooth control, IoT monitoring, and an automatic medicine dispenser; won first prize in an Embedded Systems Hackathon.

Bluetooth-Controlled RC Car *Arduino UNO, HC-05, L298N*
Built a four-motor car controlled from a smartphone app (forward, backward, left, right, and stop) using an HC-05 module and L298N motor driver.

Automatic Street Light *ESP32, IR Sensors*
Developed an automatic street light system for energy-efficient lighting and prepared an IEEE-format paper on it.

Leadership and Activities

Class Representative (CR) 2024 – Present
Department of ECE, East Point College of Engineering and Technology

- Coordinate communication between students and faculty and help resolve student concerns.

IEEE Student Member
IEEE Student Branch

- Participated in IEEE Day celebrations, technical workshops, and seminars, and supported technical events.

Achievements

- Participated in a National-Level Hackathon (Smart Health Care and Personal Safety domain).
- First Prize Winner in an Embedded Systems Hackathon.
- First Prize in the Digital Electronics Quiz Competition by Maven Silicon during the VLSI Workshop.
- Attended technical workshops and visited exhibitions at KTPO, Whitefield, Bangalore.

Declaration

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.

Place: Bangalore **Darshan Mathapati**